Computer Vision

05 Image Stitching

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Panorama

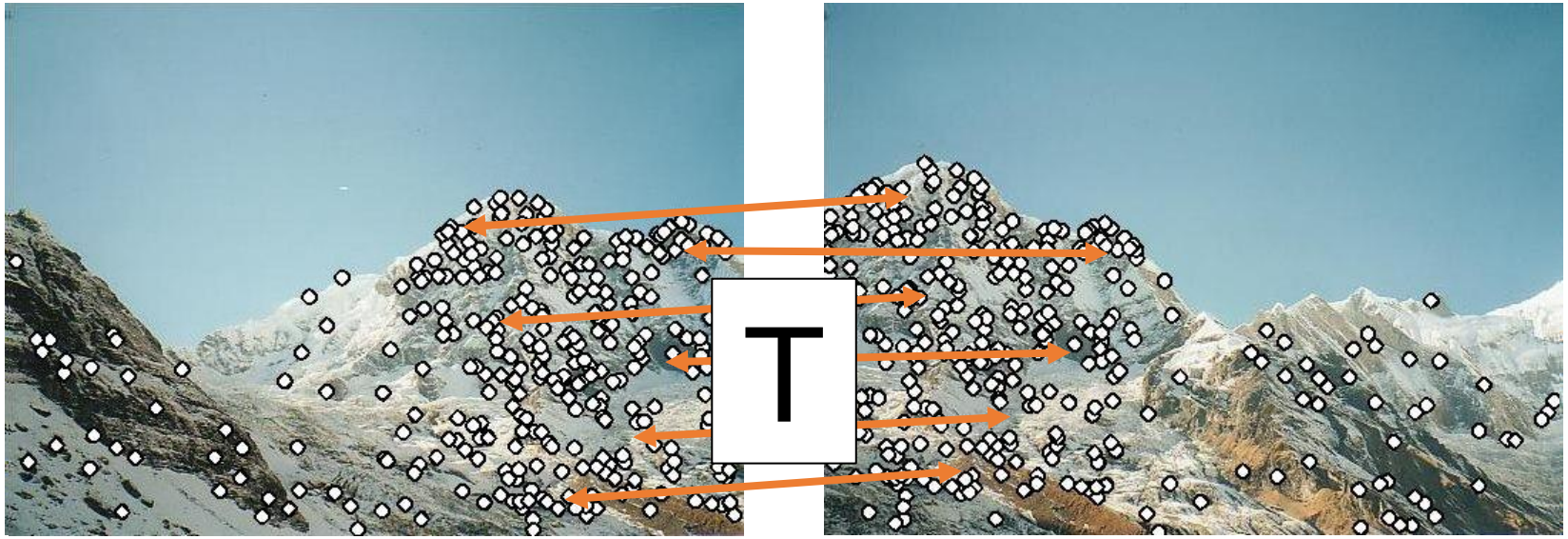


How to Build a **Panorama?**

1: Extract feature points

2: Feature matching

3: Solve transformations



How to Build a **Panorama?**

1: Extract feature points

2: Feature matching

3: Solve transformations

4: Blend Images



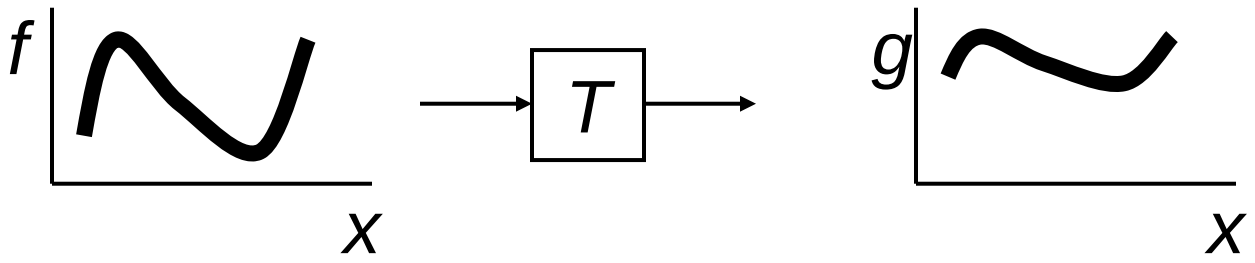
Solve Transformations



Review: Image Transformations

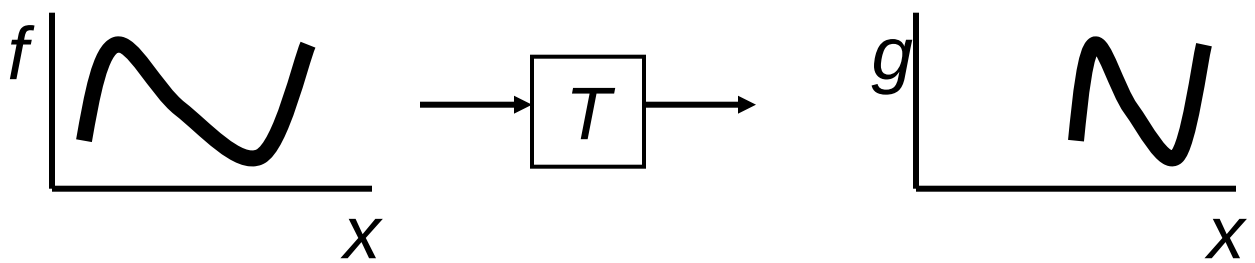
$$g(x) = \boldsymbol{T}(f(x))$$

Image filtering: change the range of images



$$g(x) = f(\boldsymbol{T}(x))$$

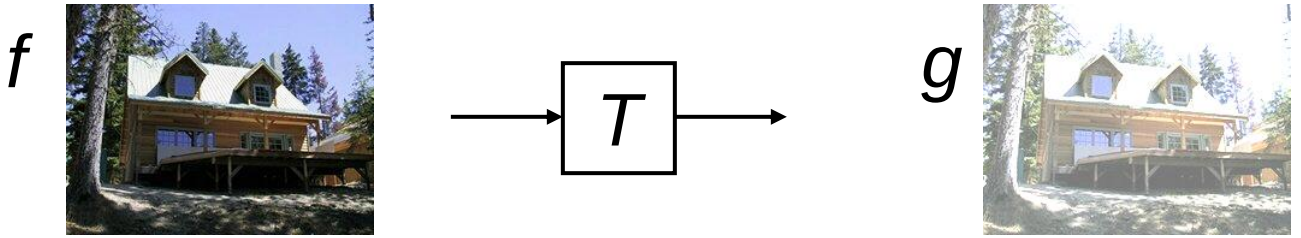
Image warping: change the ***domain*** of images



Review: Image Transformations

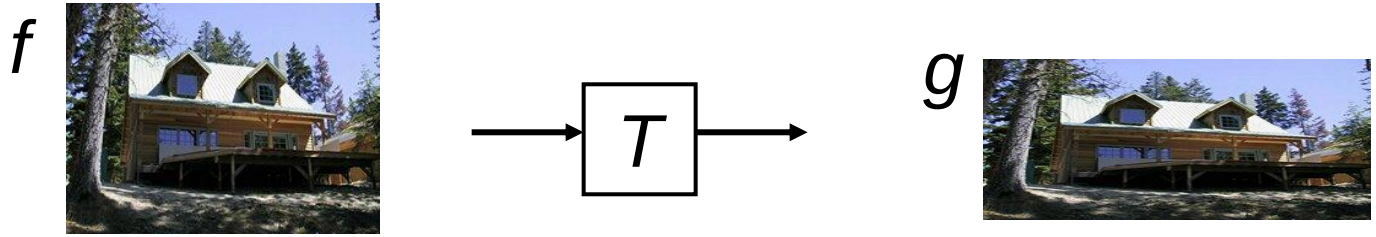
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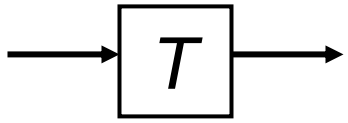
Image warping: change the ***domain*** of images



Examples of Global Image Warping

#1 Define T; #2 Solve T.





$$\boldsymbol{p}' = T(\boldsymbol{p})$$

T is the same for all points and does not depend on image contents; p and p' are 2D pixel coordinates.



translation



rotation



aspect



affine



perspective

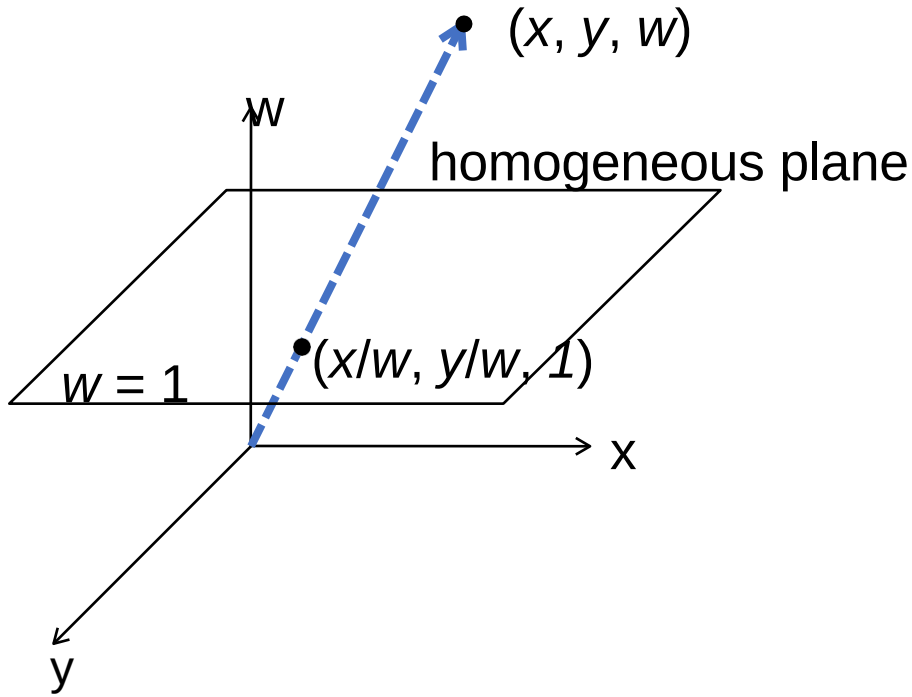


cylindrical

Transformation with Homogeneous Coordinates (#1)

Homogeneous Coordinates

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad \begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$



Linear transformation plus translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \equiv \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \equiv T \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Degree of freedom: 6

We get the follow transformation without homogeneous coordinates

$$\boldsymbol{x' = Ax + b}$$

Examples of Transformations

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \equiv \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \equiv T \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translate

Degree of freedom: 2

$$\begin{bmatrix} \boldsymbol{x}' \\ \boldsymbol{y}' \\ 1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{s}_x & 0 & 0 \\ 0 & \boldsymbol{s}_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \\ 1 \end{bmatrix}$$

Scale

Degree of freedom: 2

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2D *in-plane* rotation

Degree of freedom: 1

$$\begin{bmatrix} \boldsymbol{x}' \\ \boldsymbol{y}' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & \boldsymbol{sh}_x & 0 \\ \boldsymbol{sh}_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \\ 1 \end{bmatrix}$$

Shear

Degree of freedom: 2

Solve Transformations (#2)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \equiv \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \equiv T \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Degree of freedom: 6

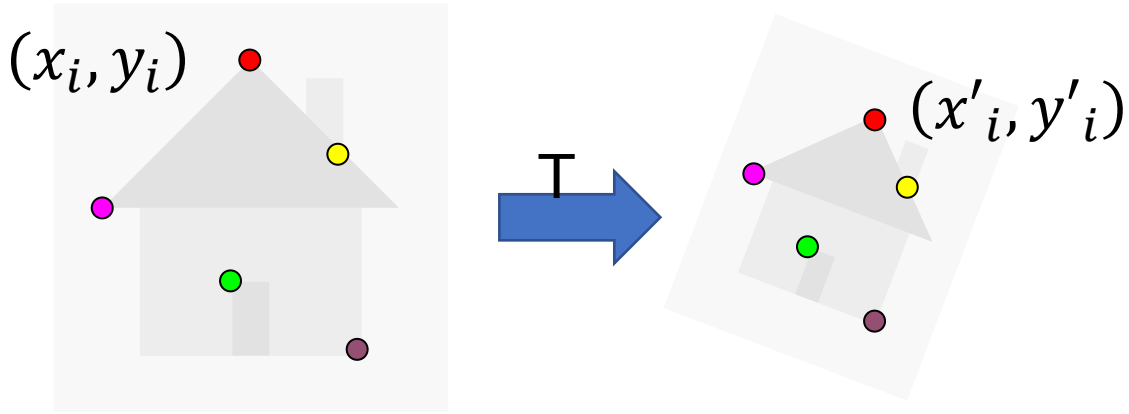
Given correspondences: $[x'_i, y'_i] \leftrightarrow [x_i, y_i]$

$$\begin{bmatrix} x_i' \\ y_i' \end{bmatrix} = \begin{bmatrix} a & b \\ d & e \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} c \\ f \end{bmatrix}$$

Set up two equations per point

$$\begin{bmatrix} \vdots \\ x_i' \\ y_i' \\ \vdots \end{bmatrix} = \begin{bmatrix} & \cdots & & & & \\ x_i & y_i & 0 & 0 & 1 & 0 \\ 0 & 0 & x_i & y_i & 0 & 1 \\ & & \cdots & & & \end{bmatrix} \begin{bmatrix} a \\ b \\ d \\ e \\ c \\ f \end{bmatrix}$$

$\mathbf{b}_{2n \times 1} = \mathbf{A}_{2n \times 6} \mathbf{t}_{6 \times 1}$



2 equations per point, 6 unknowns

How many points do we need to properly constrain the problem?

Solve Transformations (#2)

- We need 3 points (not colinear)

$$\mathbf{A}_{2n \times 6}\,\mathbf{t}_{6 \times 1} = \mathbf{b}_{2n \times 1}$$

2 equations per point, 6 unknowns

How many points do we need to properly constrain the problem?

- In real applications, we can find much more point pairs in correspondence.
- We seek the optimal values by minimizing the following energy function:

$$\boldsymbol{E} = \|\boldsymbol{At} - \boldsymbol{b}\|_2^2 = \boldsymbol{t^T A^T A t} - \boldsymbol{2 t^T A^T b} + \boldsymbol{b^T b}$$

- $\boldsymbol{E}$ is quadratic and we take its first derivative and let it to be zero:

$$\mathbf{A}^{\mathrm{T}}\mathbf{A}\mathbf{t} = \mathbf{A}^{\mathrm{T}}\mathbf{b} \qquad \mathbf{t} = \left(\mathbf{A}^{\mathrm{T}}\mathbf{A}\right)^{-1}\mathbf{A}^{\mathrm{T}}\mathbf{b}$$

Least Squares

$A^T A$ is usually of full rank! Why?

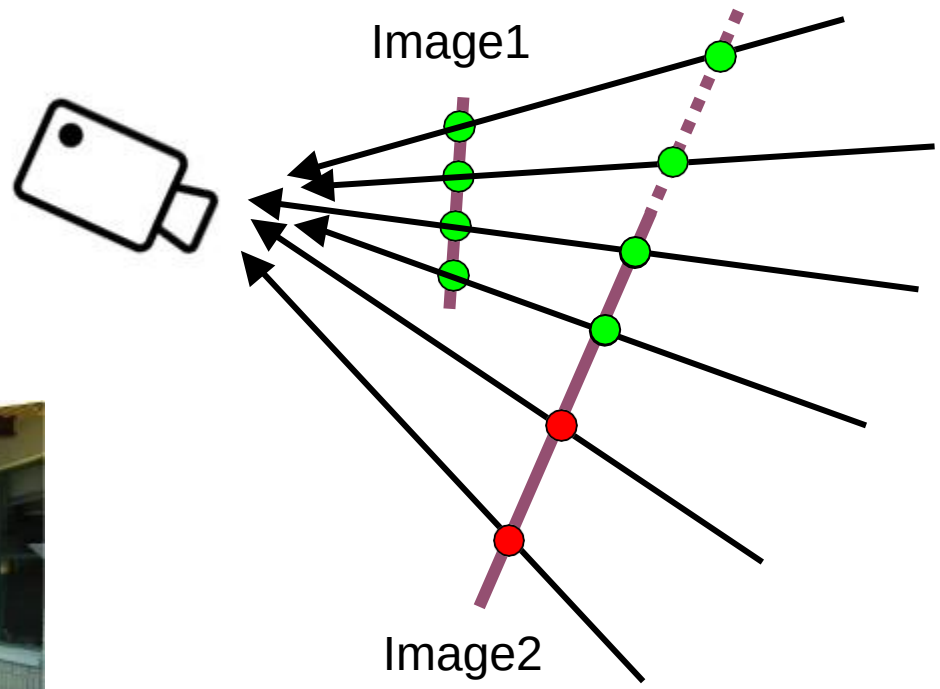
Solve Transformations (#2)







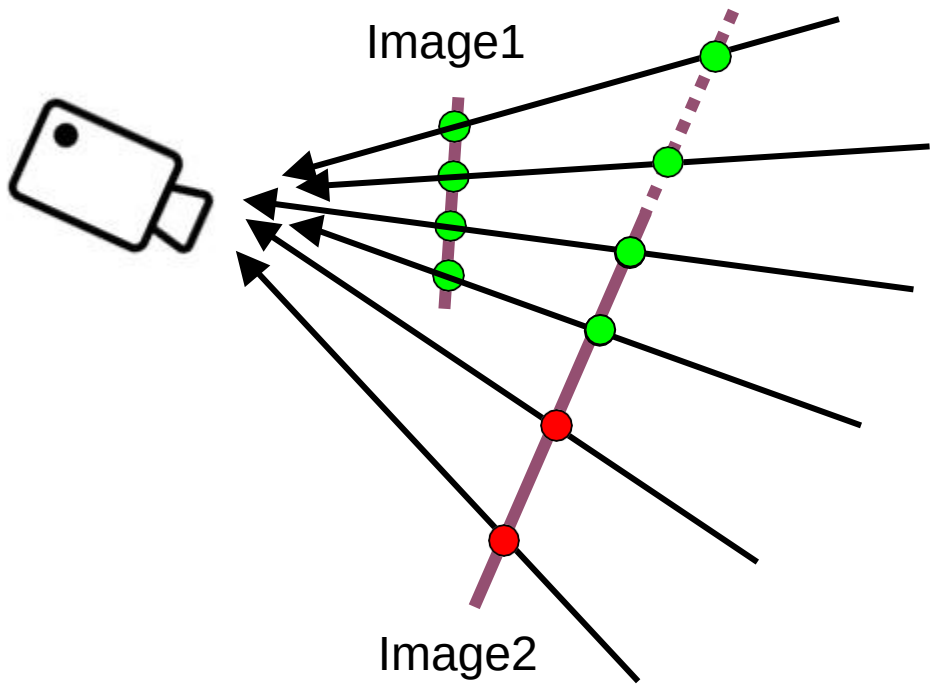
What is the relationship between these two images?



It is not affine! The following equation can not be used.

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \equiv \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \equiv T \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homography



- Homo – Same
- Homography – graphy with the same source

- Homography is a 3×3 matrix
  - The last row is $[g, h, i]$, not $[0, 0, 1]$

- Also known as projective transformation
  - Why is the last row not [0, 0, 1]?

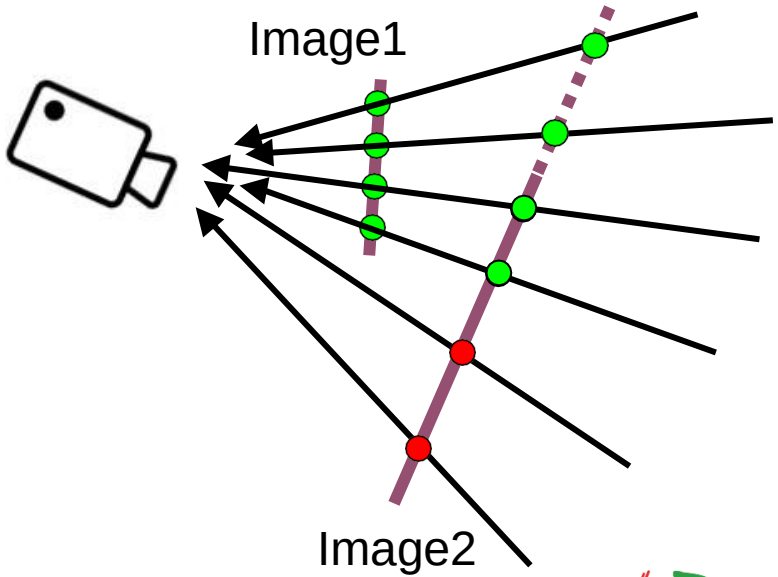
It is not affine! The following equation can not be used.

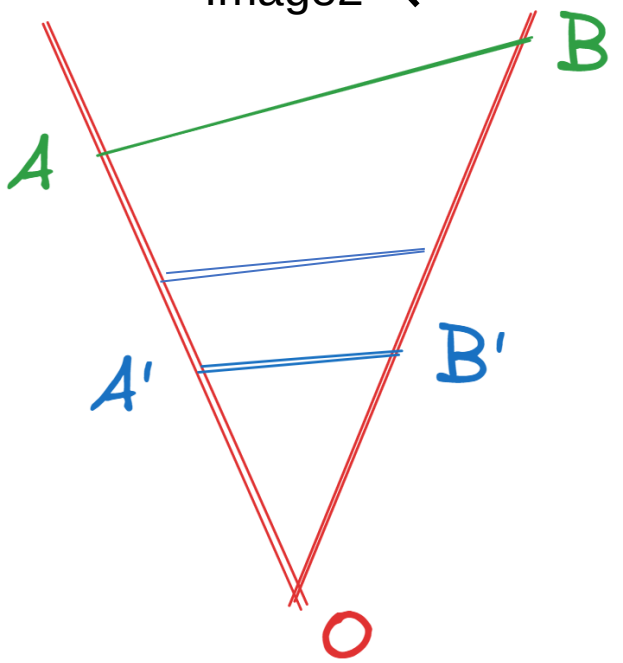
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \equiv \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \equiv T \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homography

Why is the homography matrix a 3×3 matrix?

- Transform image AB to $A'B'$, where $A'B'$ is the desired image plane.
- The transformation matrix between OAB and $OA'B'$ is apparently a full rank 3×3 matrix.
- Note that the camera center O is the shared camera center.
- If we transform AB to an image plane in parallel with $A'B'$, we get the same image. Therefore **the degree of freedom** of homography is **8**.





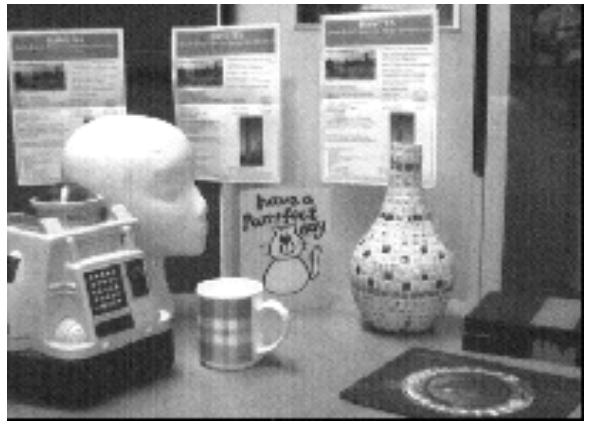
Homographies in the Real World

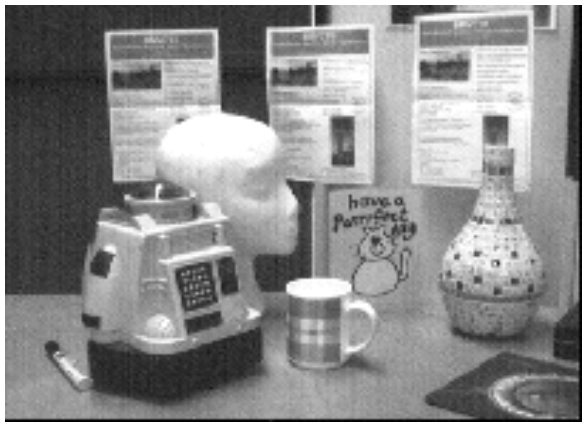
- The transformation between two views of a planar surface





- The transformation between images from two cameras that share the same center





Solve Homographies (#2)

Given correspondences: $[x'_i, y'_i] \leftrightarrow [x_i, y_i]$

Degree of freedom: 8

- Set $h_{22} = 1$
- Set the length of the vector $[h_{00}\ h_{01}\ \ldots\ h_{22}] = 1$

$$\begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} \cong \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

$$x'_i = \frac{h_{00}x_i + h_{01}y_i + h_{02}}{h_{20}x_i + h_{21}y_i + h_{22}}$$

$$y'_i = \frac{h_{10}x_i + h_{11}y_i + h_{12}}{h_{20}x_i + h_{21}y_i + h_{22}}$$

$$x'_i(h_{20}x_i + h_{21}y_i + h_{22}) = h_{00}x_i + h_{01}y_i + h_{02}$$
$$y'_i(h_{20}x_i + h_{21}y_i + h_{22}) = h_{10}x_i + h_{11}y_i + h_{12}$$

Solve Homographies (#2)

$$
\begin{aligned}
x_i'(h_{20}x_i + h_{21}y_i + h_{22}) &= h_{00}x_i + h_{01}y_i + h_{02} \\
y_i'(h_{20}x_i + h_{21}y_i + h_{22}) &= h_{10}x_i + h_{11}y_i + h_{12}
\end{aligned}
$$

$$
\begin{bmatrix}
x_1 & y_1 & 1 & 0 & 0 & 0 & -x_1'x_1 & -x_1'y_1 & -x_1' \\
0 & 0 & 0 & x_1 & y_1 & 1 & -y_1'x_1 & -y_1'y_1 & -y_1' \\
 & & & & & \vdots & & & \\
x_n & y_n & 1 & 0 & 0 & 0 & -x_n'x_n & -x_n'y_n & -x_n' \\
0 & 0 & 0 & x_n & y_n & 1 & -y_n'x_n & -y_n'y_n & -y_n'
\end{bmatrix}
\begin{bmatrix}
h_{00} \\ h_{01} \\ h_{02} \\ h_{10} \\ h_{11} \\ h_{12} \\ h_{20} \\ h_{21} \\ h_{22}
\end{bmatrix}
=
\begin{bmatrix}
0 \\ 0 \\ \vdots \\ 0 \\ 0
\end{bmatrix}
$$

$\mathbf{A}_{2n \times 9}$ $\mathbf{h}_9$ $\mathbf{0}_{2n}$

Solve Homographies (#2)

- We need 4 points (not colinear)

$$\mathbf{A}_{2n \times 9}\,\mathbf{h}_{9} = \mathbf{0}_{2n}$$

2 equations per point, 8 degree of freedom

How many points do we need to properly constrain the problem?

- In real applications, we can find much more point pairs in correspondence.
- Minimizing the following energy function by considering $\|\boldsymbol{h}\|^2 = 1$

$$\boldsymbol{E} = \|\boldsymbol{Ah}\|^2 \qquad \boldsymbol{L} = \|\boldsymbol{Ah}\|^2 + \boldsymbol{\lambda}\,(\|\boldsymbol{h}\|^2 - 1) \quad = \boldsymbol{h^T A^T A h} + \boldsymbol{\lambda h^T h} - \boldsymbol{\lambda}$$

- $\boldsymbol{L}$ is quadratic and we take its first derivative and let it to be zero:

$$\boldsymbol{A^T A h} = \boldsymbol{\lambda h}$$

$$\boldsymbol{E} = \boldsymbol{\lambda}$$

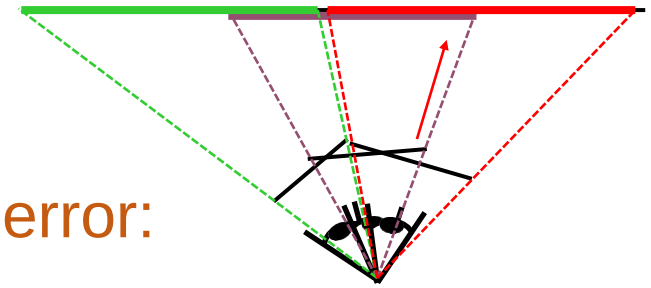
$\boldsymbol{h}$ is the Eigenvector of $\boldsymbol{A^T A}$ with the smallest eigenvalue.

Summary of Solving Homographies

Summary

- Build the matrix ***A*** according to the homography transformation.
- Compute the Eigenvector of $\boldsymbol{A^T A}$ with the smallest eigenvalue.

$$\begin{bmatrix} x_i' \\ y_i' \\ 1 \end{bmatrix} \cong \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$



Caveat

- $\|\mathbf{Ah}\|^2$ measures the *algebraic error* for optimization.
- Use the following function to measure the geometric error:

$$\sum_{i=1}^{k} \|[x_i', y_i'] - T([x_i, y_i])\|^2 + \|[x_i, y_i] - T^{-1}([x_i', y_i'])\|^2$$


Feature Matching

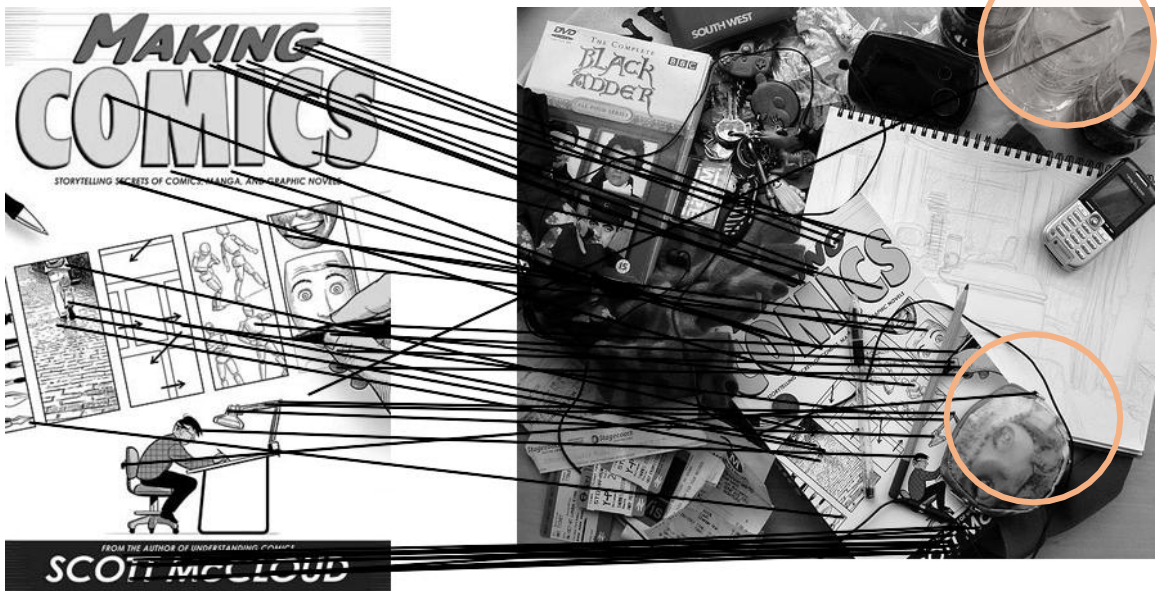
Feature Matching

Given images A and B

- Compute image features for A and B
- Match features between A and B with the nearest neighborhood searching
- Compute homography between A and B on the set of matches

- What could go wrong?

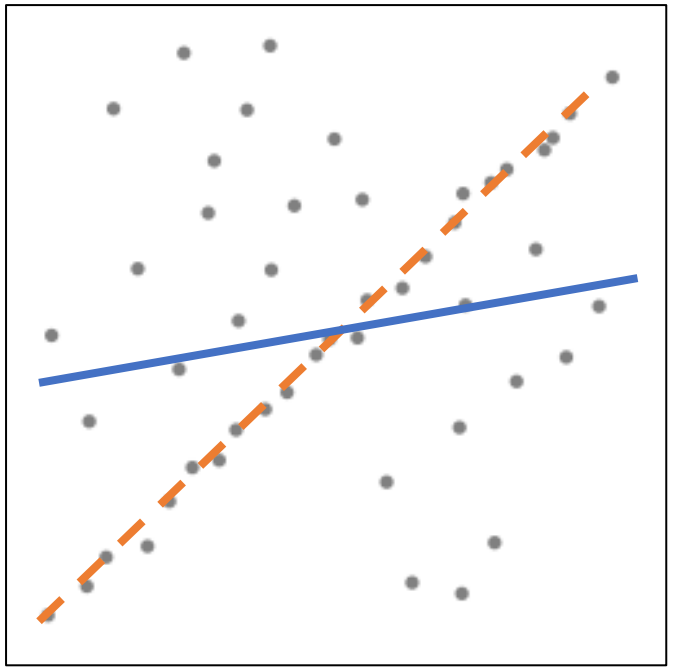
outliers



The Affect of Outliers

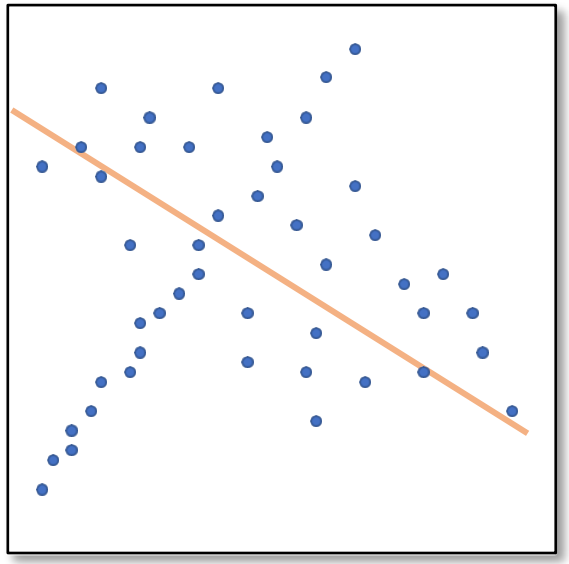
- Linear Regression
  - $y = ax + b$
  - $E = \sum_i (ax_i + b - y_i)^2$
  - Minimize E to find $a,\ b$

- If there are a lot of outliers, we get a result far from the ground truth.
- The key is to eliminate the affect of outliers

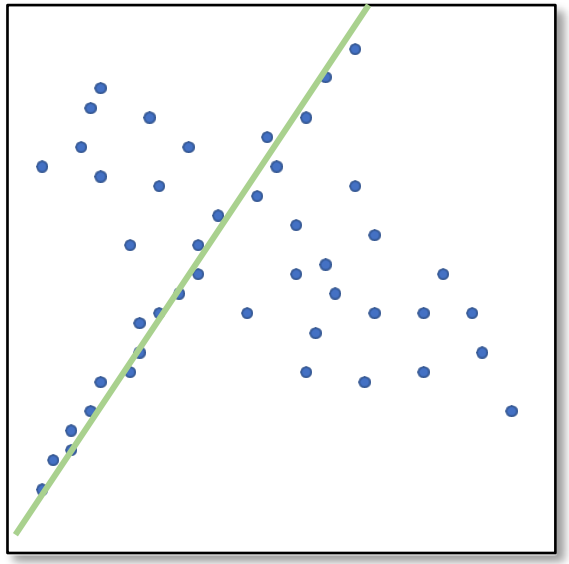


RANSAC: RANdom SAmple Consensus

- Randomly choose s samples
  - Typically s = minimum sample size that lets you fit a model
- Fit a model to those samples
  - Count the number of inliers that approximately fit the model
- Repeat N times
- Choose the model that has the largest set of inliers
- Fit the model with all inliers



Inliers: 3



Inliers: 20

RANSAC: RANdom SAmple Consensus

- How to find inliers?
  - After fitting the model, use an inlier threshold related to the amount of noise
  - We can use the geometric error of homography to measure the inliers/outliers
- When to terminate?
  - The number of iterations is related to the percentage of outliers, and the probability of success we'd like to guarantee
  - Suppose the proportion of inliers is G and the model needs P pairs to fit, the probability that we have not picked a set of inliers after N iterations: $(1 - G^P)^N$.
  - If $G = 50\%, P = 4, N = 100$, then $0.5^4 = 0.0625,\ (1 - 0.5^4)^{100} = 0.00157$
  - If $G = 30\%, P = 4, N = 100$, then $0.3^4 = 0.0081,\ (1 - 0.3^4)^{100} = 0.44339,$ $(1 - 0.3^4)^{500} = 0.017137,\ (1 - 0.3^4)^{1000} = 0.000293$
  - If we want a failure probability of at most e, then $N > \frac{\log e}{\log(1-G^P)}$.

RANSAC: RANdom SAmple Consensus

- Pros
  - Simple and general
  - Applicable to many problems
  - Often works well in practice
- Cons
  - Has parameters to tune
  - May fail for low inlier ratios
  - Requires too many iterations
- Other methods to deal with outliers
  - Robust statistics

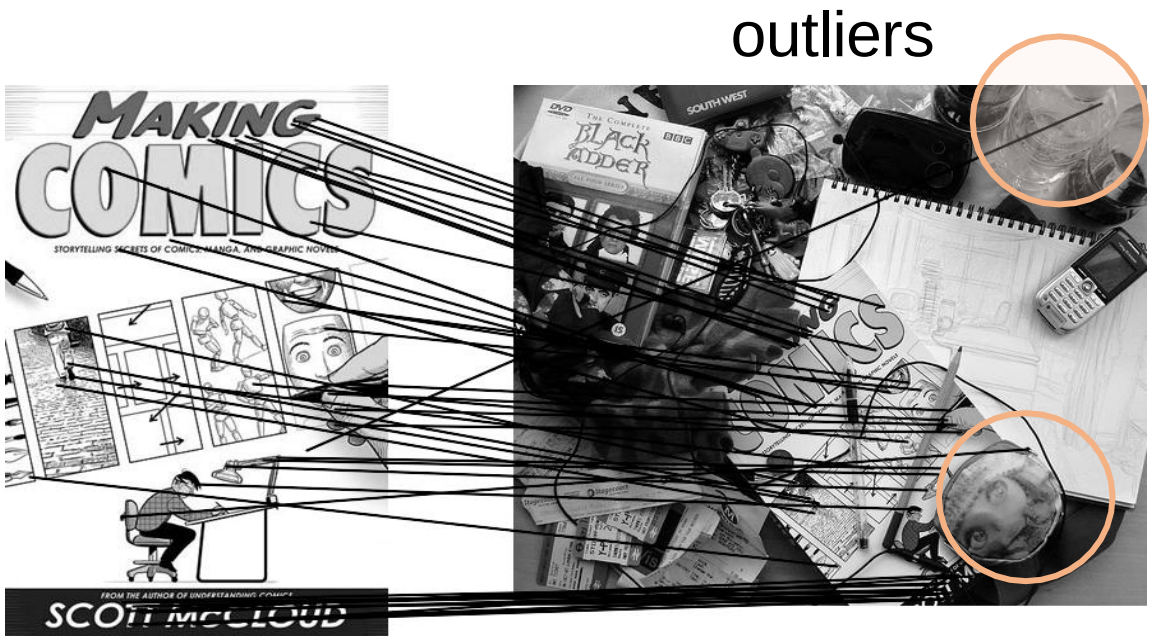
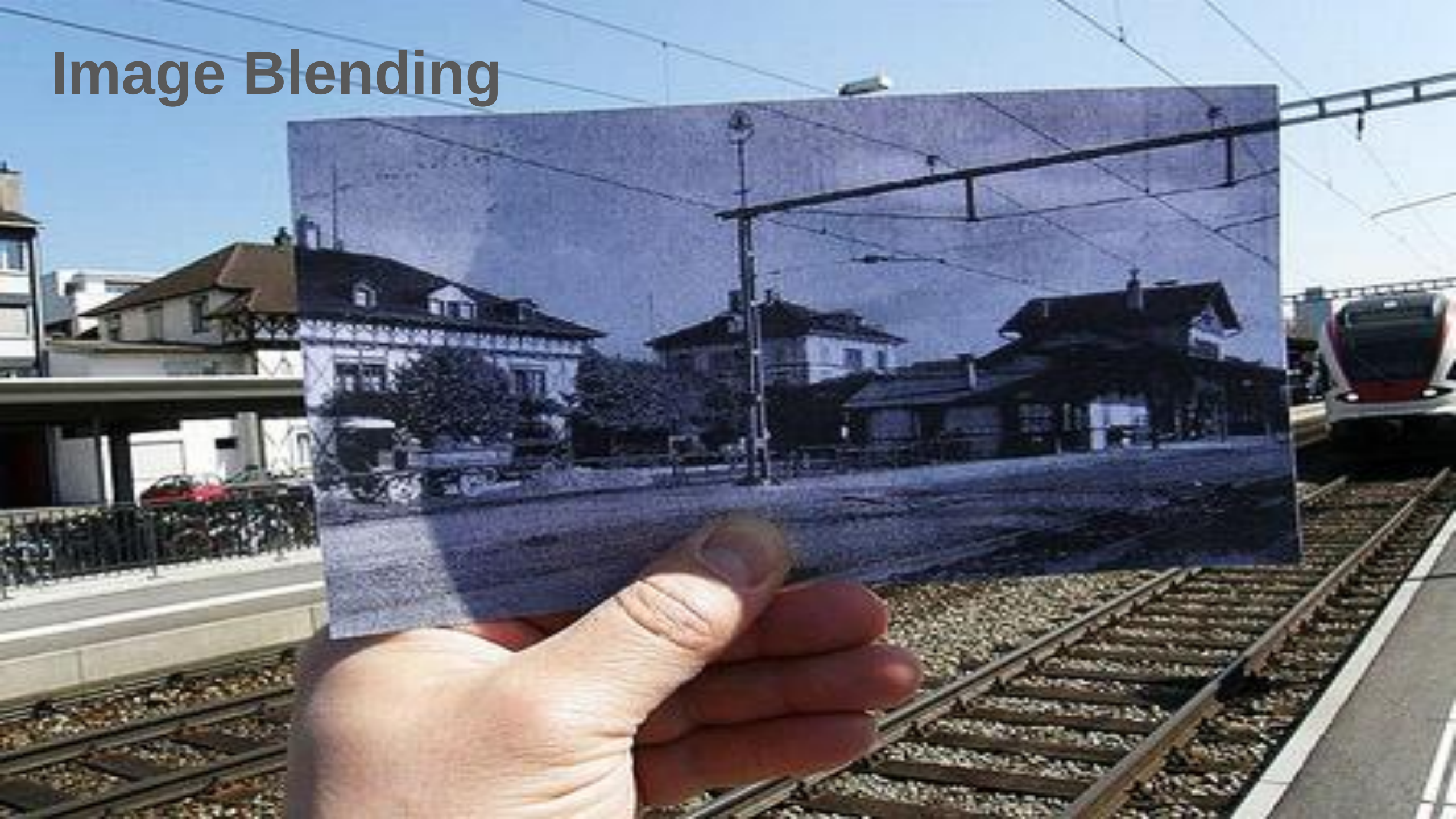


Image Blending



How to Build a **Panorama?**

1: Extract feature points

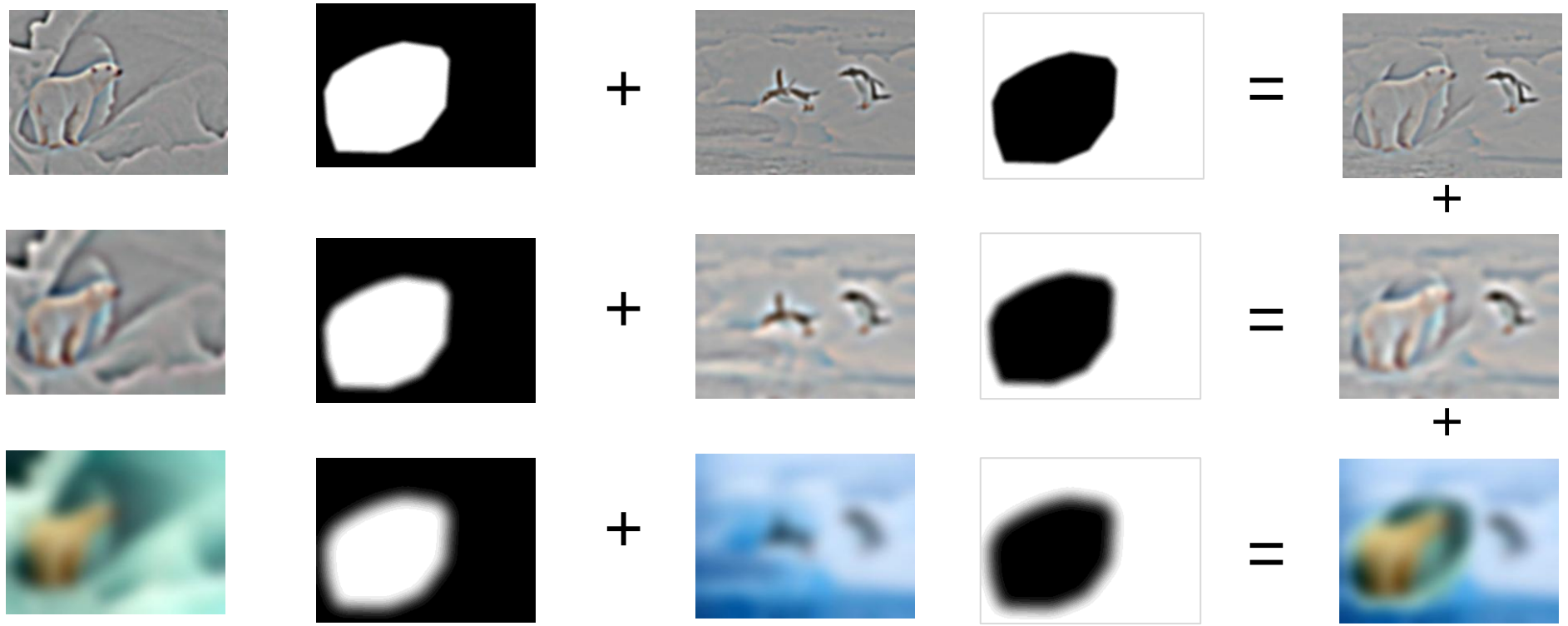
2: Feature matching

3: Solve transformations

4: Blend Images



Review: Image Blending with Laplacian Pyramid



Key Idea: blend lower frequency bands over larger spatial ranges, high frequency bands over smaller spatial ranges



Review: Image Blending with Laplacian Pyramid



- Blend high frequency bands over smaller spatial ranges



- Blend lower frequency bands over larger spatial ranges.
- The long range blending may mix contents from two image.

Poisson Image Editing

- The tones of the source images are not compatible with the target images.
- Key Idea: the gradients of the source images should be kept.

$$E = \min_f \sum_{(i,j)\in\Omega} \|\nabla f(i,j) - \nabla g(i,j)\|_2^2$$

$$f(i,j) = f^*(i,j) \;\; for \; (i,j) \in \partial\Omega$$

- where g is the source image, and f^* is the targe image, f is the output image, Ω is the region for blending.



Source Image

Target Image



cloning

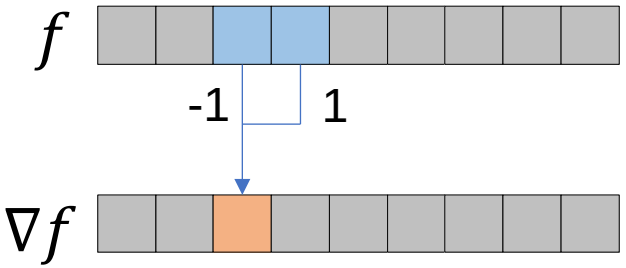
seamless cloning

Poisson Image Editing

- How to solve this optimization problem?

$$E = \min_f \sum_{(i,j)\in\Omega} \|\nabla f(i,j) - \nabla g(i,j)\|_2^2 \qquad f(i,j) = f^*(i,j) \;\; for \;(i,j) \in \partial\Omega$$

- ∇f can be written in a matrix form



$$\nabla = \begin{pmatrix} -1 & 1 & \cdots & 0 & 0 \\ 0 & -1 & & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & & -1 & 1 \\ 0 & 0 & \cdots & 0 & -1 \end{pmatrix}$$

A

B

Least Squares

$$\min_f \|Af - G\|^2 + \lambda \|Bf - F\|^2$$

$$(A^T A + \lambda B^T B) f = b$$

Large-scale Sparse Matrix

More Results of Poisson Image Editing





source

destination



Poisson Image Editing for Panorama

- Solve the homography
- Warp the source image to the reference image
- Warp the mask image to the reference image
- Run the Poisson Editing algorithm by keeping the gradient in the mask region



Summary: Build a **Panorama**

1: Extract feature points

2: Feature matching

3: Solve transformations

4: Blend Images



Reference: Chapter 8 of the *Computer Vision* book